

Assignment /

Assignment #: 1

Topic : Control Flow Statements

SELECTION

Level 1:

Easy:

1. Take values of length and breadth of a rectangle from user and check if it is square or not.

Sample Input: 5, 3

Sample Output: Not a Square

Sample Input: 7, 7

Sample Output: Square

2. Write a program to check whether the given number is odd or even.

Sample Input: 20

Sample Output: Even

Sample Input: 87

Sample Output: Odd

3. Write a program to get a number from the user and print whether it is positive, negative or zero.

Sample Input: 56

Sample Output: positive

Sample Input: -235

Sample Output: negative

4. Write a program to check whether a kid is eligible for kinder garden school admission. If a child is of age 4 or more then he/she is eligible.

Sample Input: 4

Sample Output: Eligible

Sample Input: 2

Sample Output: Not Eligible

5. Write a program to find the mobile chosen is within the budget or not. To find the budget mobiles is based on the below-mentioned criteria,
 - a) If the cost of the mobile chosen is less than or equal to 15000 then display it as "Mobile chosen is within the budget"
 - b) If the cost of the mobile chosen is greater than 15000 then display it as "Mobile chosen is beyond the budget"

Sample Input: 12000

Sample Output: Mobile chosen is within the budget

Sample Input: 22000

Sample Output: Mobile chosen is beyond the budget.

Level 2: Medium

1. There is a movie club for under 20 in the city where the entry is age restricted. Based on their age allow them into either 'Cartoon Club' or 'Teens Club'.

Sample Input: 10

Sample Output: Cartoon Club

Sample Input: 15

Sample Output: Teens Club

Sample Input: -10

Sample Output: Invalid Age

Sample Input: 25

Sample Output: Not Allowed

2. Write a program if an integer variable currentNumber is odd, change its value so that it is now 3 times currentNumber plus 1, otherwise change its value so that it is now half of currentNumber.

Sample Input: 17

Sample Output: 52

Sample Input: 26

Sample Output: 13

3. Write a program to read an integer variable 'Code'. If the Code value is 1, read double values and print the sum. If the Code value is 2, read the integers and print the product. If the code value is 3, read the strings and join them.

Sample Input: 1, 24.50, 67.00

Sample Output: 91.5

Sample Input: 3, 'Hello', 'World'

Sample Output: HelloWorld

4. Once a baby lion lost his way in the jungle. An old deer took pity on him and planned to take him to his place. But the other deer and his other friends — rabbits, squirrels, and birds are scared so they accompanied. In the morning they counted themselves to see if the baby lion done any mischief. Help them to find it. Total number of animals, count of each (rabbit, deer, birds, and squirrels) in the morning are the inputs.

Sample Input: 240, 27,48,124,38

Sample Output: Baby lion is mischievous

Sample Input: 250, 42,46,115,47

Sample Output: Baby lion is well behaved

Sample Input: 120, 45,38,30, 27

Sample Output: Counted wrongly



5. A store charges \$12 per item if you buy less than 10 items. If you buy between 10 and 99 items, the cost is \$10 per item. If you buy 100 or more items, the cost is \$7 per item. Write the logic that asks customer name, how many items they are buying and prints the customer's name and total cost.

Sample Input: Smith, 40

Sample Output: Smith 400

Sample Input: Eve, 111

Sample Output: Eve 777

Level 3: Hard

1. A student will not be allowed to sit in exam if his/her attendance is less than 75%. Number of classes held, and the Number of classes attended are the inputs. Display the attendance percentage and the eligibility of the student for the exam. Allow the student to sit if he/she has

medical cause. Ask user if he/she has medical cause or not ('Y' or 'N') only when the attendance is lacking and print accordingly.

Sample Input: 100, 80

Sample Output: 80% Allowed

Sample Input: 100, 60, 'N'

Sample Output: 60% Not allowed

Sample Input: 100, 70, 'Y'

Sample Output: 70% Allowed

2. Jessica teaches her students about how many days in a month, what is a leap year and how to find it. Write a program to help her by saying the answer (number of days in a month) to expect from the student for the given month number and the year.

Sample Input: 2, 2016

Sample Output: February 2016 has 29 days

Sample Input: 7, 2020

Sample Output: July 2020 has 31 days

Sample Input: 2, 2019

Sample Output: February 2019 has 28 days

3. Riya's teacher has asked her to prepare well for the lesson on seasons. When her teacher tells a month, she needs to say the season corresponding to that month. Write a program to solve the above task. Spring - March to May, Summer - June to August, Autumn - September to November and, Winter - December to February. Month should be in the range 1 to 12. If not, the output should be "Invalid month".

Sample Input: 11

Sample Output: Autumn

Sample Input: 13

Sample Output: Invalid month

4. A person pays tax 10% if his income is between 2.5 to 5 lacs. On the amount greater than that 20% for 5 to 12 lacs, for amount above 12 lacs 30%. Person will enter the amount input and will get tax amount as output.

Sample Input: 200000

Sample Output: You are exempted from tax

Sample Input: 40000

Sample Output: Tax amount is 15000

Sample Input: 800000

Sample Output: 85000

Sample Input: 1300000

Sample Output: 195000

Iteration

Easy:

1. Write a program to print the first N natural numbers if N is the input.

Sample Input: 5

Sample Output: 1 2 3 4 5

Sample Input: 7

Sample Output: 1 2 3 4 5 6 7

2. Write a program to print the power of m raised to n.

Sample Input: 7, 5

Sample Output: 16807

Sample Input: 4, 6

Sample Output: 4096

3. Kartik wants to help his younger brother in mathematics homework. Brother will enter the number and output should give table of that number. Write a program to print table.

Sample Input: 4

Sample Output: 4 8 12 16 20 24 28 32 36 40

Sample Input: 6

Sample Output: 6 12 18 24 30 36 42 48 54 60

- Smith and John, grade 3 students playing number games. When Smith gives a number John will say the natural numbers up to that number but in reverse order. Shia, their friend will say the sum of those numbers. Help John and Smith with your program.

Sample Input: 4

Sample Output: 4 3 2 1 , 10

Sample Input: 7

Sample Output: 7 6 5 4 3 2 1 , 28

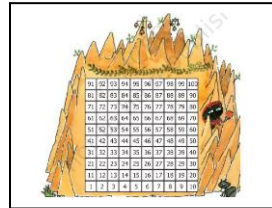
- Kittu the 'Giant Ant' has 100 rooms in its colony. All the ants in the colony can crawl from 1 room to another, from there to another and so on. But our 'Kittu' has a special power that makes him to jump from one room to next 10th room directly. That is room no 1 to room no 11 or 3 to 13... But poor 'Kittu' don't know to calculate/find which room that he can jump next. Help him saying all the room numbers to climb up. His starting room number can be the input.

Sample Input: 2

Sample output: 2 12 22 32 ... 92

Sample Input: 5

Sample Output: 5 15 25 35 ... 95



Medium:

- Write a program to read a number N. Find the sum of odd and even numbers from 1 to N numbers (inclusive).

Sample Input: 10

Sample Output: 25 30

Sample Input: 15

Sample Output: 64 56

- Reina and Sierra were playing a game. Reina would give out number, and Sierra must reverse the given number. Help Sierra by writing program to reverse the number. Note that Reina should give a five-digit number.

Sample Input: 61987

Sample Output: 78916

Sample Input: 1234

Sample Output: 'Not a valid number'

3. Write a program to check whether the given number is prime or not.

Sample Input: 91

Sample Output: No

Sample Input: 97

Sample Output: Yes

4. Write a program to print all prime numbers between 1 and N(inclusive). N will be the input.

Sample Input: 5

Sample Output: 2 3 5

5. Write a program to find the sum of the even and odd digits of the number which is given as input.

Sample Input: 5672

Sample Output: 8 12

Sample Input: 97641

Sample Output: 16 11

Hard:

1. Given a start number and end number, you need to display the numbers inclusive but “ding” for every 5th number and “dong” for every 10th number.

Inputs: Starting number and Ending number

Outputs: Display “ding” for every 5th number and “dong” for every 10th number

Sample Input: 11, 40

Sample Output: 11 12 13 14 ding 16 17 18 19 dong 21 22 23 24 ding 26 27 28 29 dong 31 32 33 34
ding 36 37 38 39 dong

2. Write a program to find whether the given number of any digit is an Armstrong number or not.

Sample Input: 1634

Sample Output: true

Sample Input: 22121

Sample Output: false

3. Write a program that reads the prices of a shop's products continuously until the user enters -1. The program should display the minimum price, the maximum, and the average of those within [5,30], before it terminates. Assume that none of the product costs more than \$100.

Sample Input: 20 10 2 8 3 9 12 15 60 80 25 20 -1

Sample Output: 80 2 14

4. Write a C# program to check whether the pair is amicable pair.

Amicable Pair

The sum of all proper divisor of x and the sum of all proper divisor y should be equal to the opposite number.

Example: x = 220, y = 284

Proper divisors of 220 are 1, 2, 4, 5, 10, 11, 20, 22, 44, 55 and 110 with sum 284

Proper divisors of 284 are 1, 2, 4, 71 and 142 with sum 220.

Sample Input: 220, 284

Sample Output: true

Sample Input: 132, 182

Sample Output: false

5. Arjun wanted to find the prime factors of the given number. Help him by writing a program to find the prime factors of the given number.

Sample Input: 315

Sample Output: 3 3 5 7

Sample Input: 210

Sample Output: 2 3 5 7